GGG-L-350
MARCH 30, 1956
SUPERSEDING
Int. Fed. Spec. GGG-L-00350 (Navy-Ships)
17 November 1954

FEDERAL SPECIFICATION

LIFTER, VALVE SPRING; COMPRESSOR, VALVE SPRING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers valve rpring lifters and compressors used for removing and replacing valve springs of gasoline engines.

1.1.1 ***Federal specification coverage.*** **— Federal specifications do not include all types and classes of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover only those generally used by the Federal Government.**

1.2 Classification.

1.2.1 ***Types, classes, and styles.***—**Lifters and compressors covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):**

Type I.—Lifter.
- Class 1.—C-frame.
  - Style A.—With depth adjustment.
  - Style B.—Without depth adjustment.
- Class 2.—Hinged lever.
  - Style A.—Screw-actuated.
  - Style B.—Lever-actuated.

Type II.—Compressor, hinged lever.
- Style A.—Screw-actuated.
- Style B.—Lever-actuated.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

QQ-M-151—Metals; General Specification for inspection of.

Military Specification:

MIL-P-15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

362249—56—No. 240

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. — The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of machines which are otherwise in accordance with the requirements of this specification.

3.2 Material. — The material as specified herein for each type, class, and style of lifter and compressor shall possess such qualities as to be suitable for the purpose intended.

3.3 Construction.—Lifters and compressors shall be assembled with bolts and nuts or rivets, except parts which may be subject to disassembly during the life of the tool shall be assembled with bolts or screws.

3.4 Operation. — All parts of lifters and compressors shall operate freely without binding or binding during spring lifting and compressing operations.

3.5 Test loads. — Lifters and compressors shall withstand the test loads specified in the applicable tables without injury or permanent deformation which may affect durability or serviceability when tested as specified in 4.4.1.

3.6 Interchangeability of parts.—All parts of lifters or compressors of the same manufacture which may be subject to replacement during the normal life of the tool shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification (see 6.2.1).

3.7 Finish.—Components of lifters and compressors not of corrosion resistant material shall be provided with one of the following coatings for preventing corrosion under ordinary service conditions:

(a) Zinc plate.
(b) Cadmium plate.
(c) Chemically produced oxide.

3.8 Identification marking. — Lifters and compressors shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.9 Type I, lifter.—Lifters shall be designed for compressing valve springs between the jaws of the lifter and the engine block or head.

3.9.1 *Class 1, C-frame.*—Lifters shall consist essentially of a C-shaped frame, a rack and pinion gear or screw, or actuating lever or levers, an operating handle, and jaws. Lifters shall be suitable for removing and replacing valve springs of L-head, V, and overhead valve engines.

3.9.1.1 ***Material.*** — The frame, jaws, rack, and pinion gear or screw shall be of steel. Other components shall be of steel or other materials suitable for the purpose intended.

3.9.1.2 *Jaws.*—Lifters shall be provided with one pair of straight jaws, one pair of jaws suitable for General Motors and Continental engines, and one set of special offset jaws, as specified (see 6.2), for engines requiring additional offset. The jaws shall be flanged or provided with other means that will retain the spring in a normal position on the jaws. The jaws shall be heat-treated to possess a Rockwell C hardness of not less than 33 nor more than 43 when tested as specified in 4.4.2.

3.9.1.3 ***Style A, with depth adjustment.*** — Lifters shall be provided with a means for setting and maintaining the required distance between the jaws and the end of the rack or screw.

3.9.1.3.1 *Operation.*—The spring lifting operation shall be facilitated by setting the distance between the jaws and the end of the rack or screw when lifting the first spring. With this

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

setting established, the subsequent springs shall be lifted by actuating the operating handle. The lifter shall lock when it has raised the spring to the desired height.

3.9.1.3.2 Style A lifters shall conform to table I and shall be similar to figure 1.

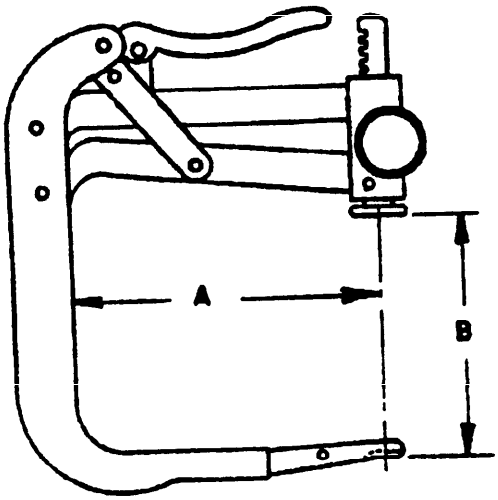


FIGURE 1.—Type I, class 1, style A lifter, C-frame, with depth adjustment.

3.9.1.4 *Style B, without depth adjustment.*—Lifters shall be so designed that valve springs may be compressed and inserted into valve spring holders to hold the compressed spring while it is being placed into position in the engine. When specified (see 6.2), a set of three valve spring holders, similar to figure 3, shall be furnished with each style B lifter.

3.9.1.4.1 *Operation.*—The spring lifting operation shall be facilitated by turning the operating handle. The lifter shall lock automatically at any desired height. The lifted spring shall not return to its normal position until the locking device is released.

3.9.1.4.2 Style B lifters shall conform to table II and shall be similar to figure 2.

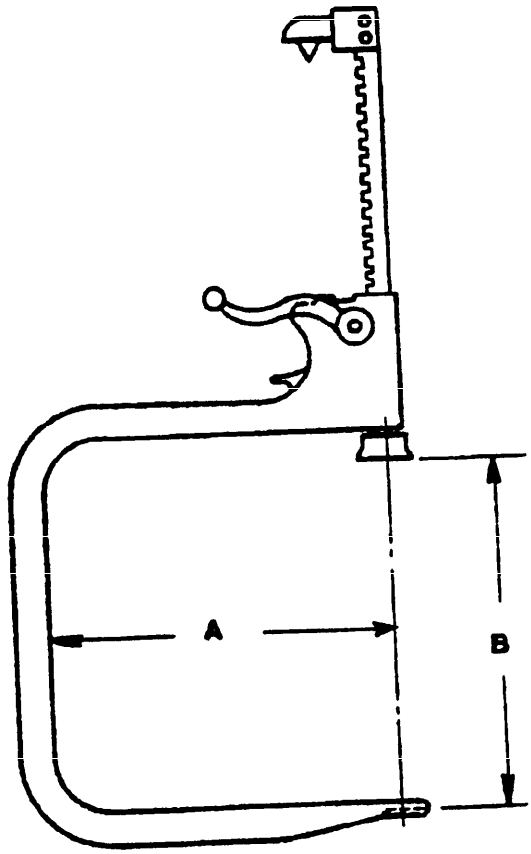


FIGURE 2.—Type I, class 1, style B lifter, C-frame, without depth adjustment.

TABLE I.—*Type I, class 1, style A lifter, C-frame, with depth adjustment*

Size number	Capacity (valve spring outside diameter)	A		B (full open position)		Weight (approx.)	Test load (min.)
		Min.	Max.	Min.	Max.		
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*
1	1 to 2	9	9½	10	10½	6	450
2	½ to 1½	9¾	10¼	10	10½	7¼	450
3	1 to 2	10¼	10¼	11½	12¼	8½	450

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE II.—*Type I, class 1, style B lifter, C-frame, without depth adjustment*

Size number	Capacity (valve spring outside diameter)	A		B (full open position)		Weight (approx.)	Test load (min.)
		Min.	Max.	Min.	Max.		
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*
1	1⅝	8¾	9½	10¾	11¼	5¼	350
2	1⅝	10¾	11¼	10¾	11¼	6½	400
3	1⅝	10¾	11¼	12¾	13¼	7½	450

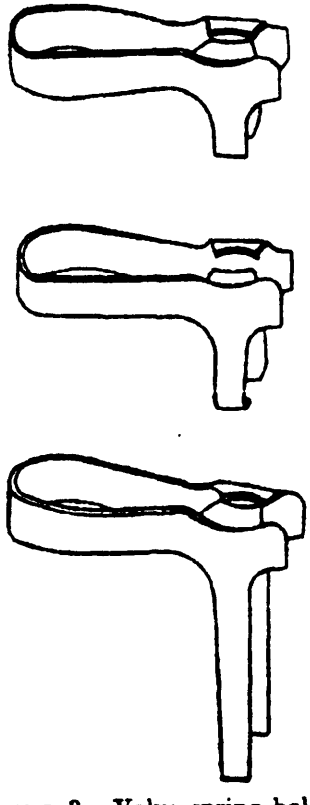


FIGURE 3.—Valve spring holders.

3.9.2 *Class 2, hinged lever.*—Lifters shall be suitable for removing and replacing valve springs of L-head gasoline engines.

3.9.2.1 *Jaws.*—The lifter shall be designed so that the jaws remain parallel to each other in all positions within the capacity of the tool. The upper jaw shall be flanged or provided with other means that will retain the spring equally on both arms of the jaw and shall be heat-treated to possess a Rockwell C hardness of not less than 33 nor more than 43 when tested as specified in 4.4.2. The lower jaw shall be designed to provide a base for the lifter. Thumbscrews shall be provided to facilitate the adjustment of the required width of the jaws. There shall be no indications of excessive looseness of the jaws when the jaws are adjusted to the required width.

3.9.2.2 *Style A, screw-actuated.*—Each style A lifter shall consist essentially of upper and lower jaws, fulcrum links, handwheel and adjusting screws.

3.9.2.2.1 *Material.*—Components of lifters shall be of steel, except the handwheel may be of any metallic material.

3.9.2.2.2 *Operation.*—The spring lifting operation shall be facilitated by turning the handwheel in a clockwise direction. The lifter shall lock automatically at any desired height and shall remain in place without additional support.

3.9.2.2.3 Style A lifters shall conform to table III and shall be similar to figure 4.

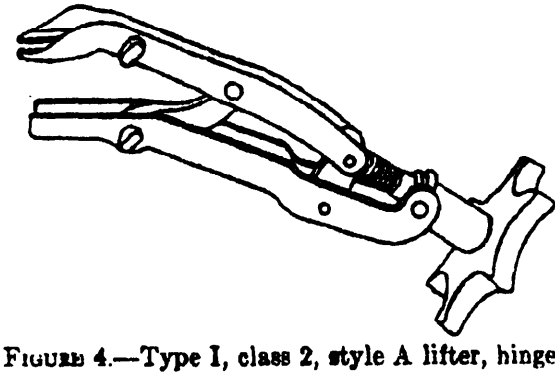


FIGURE 4.—Type I, class 2, style A lifter, hinged lever, screw-actuated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE III.—*Type I, class 2, style A lifter, hinged lever, screw-actuated*

Size number	Capacity (valve spring outside diameter)		Lift (min.)	Over-all length		Weight (max.)	Test load (min.)
	Min.	Max.		Min.	Max.		
	Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*
1	⅛	1¼	2	7½	9½	1¼	425

3.9.2.3 *Style B, lever-actuated.*—Each style B lifter shall consist essentially of two handles, two jaws, and a locking device.

3.9.2.3.1 *Material.*—Components of lifters shall be of steel.

3.9.2.3.2 *Operation.*—The spring lifting operation shall be facilitated by compressing the handles. The lifter shall lock automatically at any desired height and shall remain in place without additional support. The locking device shall be released with a minimum of effort.

3.9.2.3.3 Style B lifters shall conform to table IV and shall be similar to figure 5.

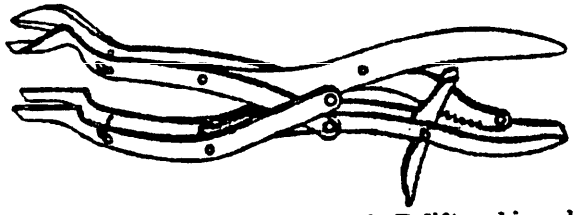


FIGURE 5.—Type I, class 2, style B lifter, hinged lever, lever-actuated.

3.10 Type II, compressor, hinged lever.—Compressors shall be suitable for removing and replacing valve springs.

3.10.1 *Material.*—Components of compressors shall be of steel.

3.10.2 *Style A, screw-actuated.*—Compressor shall consist essentially of a valve spring compressor suitable for removing and replacing valve springs for small engines such as Briggs-Stratton and Clinton. The compressor shall be provided with a valve keeper inserter similar to figure 6.

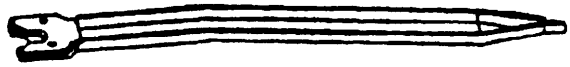


FIGURE 6.—Valve keeper inserter.

3.10.2.1 *Compressor jaws.*—The compressor jaws shall be designed to operate in small valve spring ports on engines without alteration. Both jaws shall be adjustable laterally to seat properly on top and bottom of spring. The upper jaw shall be equipped with an adjustable stop to enable the spring to be cocked at an angle when required for the removal of the spring. The par-

TABLE IV.—*Type I, class 2, style B lifter, hinged lever, lever-actuated*

Size number	Capacity (valve spring outside diameter)		Lift (min.)	Over-all length		Weight (max.)	Test load (min.)
	Min.	Max.		Min.	Max.		
	Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Pound*	*Pounds*
1	⅛	1¼	2	9	10½	1	300

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

TABLE V.—*Type II, style A, compressor, hinged lever, screw-actuated*

Size number	Capacity (valve spring outside diameter)		Compresses from	Over-all length		Weight	Test load (min.)
	Min.	Max.		Min.	Max.		
	Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*
1	⅛	1⅛	2⅛ to ⅛	9¼	10	1⅛	300

allel jaws shall be raised and lowered through the medium of a screw and toggle actuated by a handwheel at the end of the compressor opposite the jaws.

3.10.2.2 *Valve keeper inserter.* — The valve keeper inserter shall be suitable for inserting keys and pins in the valve stem. The inserter at the valve key end shall not exceed ⅝ inch in width and 5/32 inch in thickness. The keys shall be secured to the inserter by a magnet. The opposite end of the inserter shall be designed to hold and automatically center pin type locks.

3.10.2.3 Style A screw-actuated compressors shall conform to table V and shall be similar to figure 7.

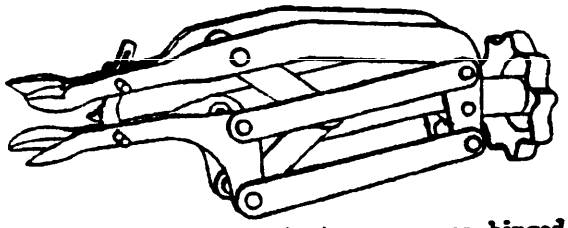


FIGURE 7.—Type II, style A compressor, hinged lever, screw-actuated.

3.10.3 *Style B, lever-actuated.*—Compressors shall be suitable for removing and replacing valve springs of overhead valve engines and replacing springs in gasoline engines of all types. Each compressor shall consist essentially of two handles, two jaws, and a locking device.

3.10.3.1 *Operation.* — Springs shall be compressed between the jaws by compressing the handles of the tool. The compressor shall lock automatically at any desired jaw position. The locking device shall be released with a minimum amount of effort.

3.10.3.2 *Jaws.*—The compressor shall be designed so that parallelism is maintained between the jaws in all positions within the capacity of the tool. The jaws shall be provided with flanges that flare out to serve as guides to bring the spring to the center of the jaws.

3.10.3.3 Style B lever-actuated compressors shall conform to table VI and shall be similar to figure 8.

TABLE VI.—*Type II, style B compressor, hinged lever, lever-actuated*

Size number	Capacity (valve spring outside diameter)		Compresses from	Over-all length		Weight (max.)	Test load (min.)
	Min.	Max.		Min.	Max.		
	Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*
1	1	1½	3 to 1	11½	12½	1⅛	250

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

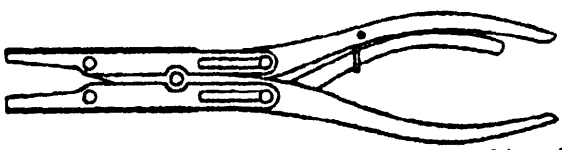


FIGURE 8.—Type II, style B compressor, hinged lever, lever-actuated.

3.11 **Workmanship.** — The workmanship shall be first class in every respect. Lifters and compressors shall be free from burrs, sharp edges, and defects which may impair durability and serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.* — All lifters and compressors of the same type, class, style and size presented at one time shall be considered a lot for purposes of inspection and tests.

4.1.2 *Sampling for visual and dimensional inspection.*—A random sample of lifters or compressors shall be taken by the Government inspector in accordance with table VII from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with Military Standard MIL-STD-105.

TABLE VII.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

Number of lifters or compressors in lot	Number of lifters or compressors in sample
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

NOTE.—The Government inspector may institute tightened or reduced inspection in accordance with section 9 of Military Standard MIL-STD-105.

4.1.3 *Sampling for lot acceptance tests.* — A random sample of lifters or compressors shall be taken by the Government inspector from each inspection lot in accordance with table VIII and shall be subjected to the tests specified in 4.3.

TABLE VIII.—*Sampling for lot acceptance tests*

Number of lifters or compressors in the lot	Number of lifters or compressors in the sample	Acceptance number	Rejection number
		(Number of items nonconforming on any lot acceptance test)	
15 and under	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 to 800	25	2	3
801 and over	35	3	4

4.2 Inspection.

4.2.1 *Visual and dimensional inspection.* — Each of the sample lifters and compressors taken in accordance with table VII shall be visually and dimensionally inspected by or under the supervision of the Government inspector to verify compliance with this specification. Any lifter or compressor in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 **Lot acceptance tests.**—Each of the sample lifters or compressors taken in accordance with table VIII shall be tested in accordance with 4.4, to verify compliance with this specification. Any sample lifter or compressor which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Test loads.* — Sample lifters and compressors shall be tested for conformance with 3.5 by applying the test load specified in the applicable table for each type, class, and style lifter or compressor. The test load shall be made by application of dead weights, tension machine or other similar device.

Source: https://assist.dla.mil — Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

4.4.1.1 *Type I, class 1 lifters.*—The test load shall be applied between the jaws and end of rack or screw in line with the axis of the direction of motion (axis of screw or rack), the direction to be such as to cause the lifter to open.

4.4.1.2 *Type I, class 2 lifters.*—The test load shall be applied to the upper and lower jaws, the direction to be such as to cause the jaws to come together.

4.4.1.3 *Type II, compressors.*—The test load shall be applied between the jaws, the direction to be such as to cause the compressor to open.

4.4.1.4 *Test failure.*—Any lifter or compressor which fails under the test load or remains permanently deformed or damaged in any manner after the removal of the test load, shall be considered as having failed the load test.

4.4.2 *Hardness test.*—The jaws of type I, class 1 lifters and the upper jaws of type I, class 2 lifters shall be tested to determine compliance with 3.9.1.2 and 3.9.2.1. Hardness tests shall be conducted in accordance with Federal Specification QQ-M-151.

4.4.3 *Rejected lots.* — Rejected lots may be offered again for Government inspection and tests provided the contractor has repaired or removed all nonconforming lifters or compressors. The Government inspector shall again take and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Nonmilitary requirements.

5.1.1 *Preservation.*—Cleaning and preservation shall be in accordance with the manufacturer's standard practice.

5.1.2 *Packaging.*—Commercial packages will be acceptable.

5.1.3 *Packing.* — The equipment shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose in such manner as to insure acceptance and safe delivery to the designated point by common or other carrier at the lowest rate of the carrier.

5.1.4 *Marking.*

5.1.4.1 *Interior containers.* — Packages and interior containers shall be marked with the name of the material and the name of the contractor, or trade-mark of such known character that the source of supply may be readily determined.

5.1.4.2 *Exterior containers.* — Shipping containers shall be marked with the name of the item, the type and class, and the quantity therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

5.2 Military requirements.

5.2.1 *Preparation for shipment.* — Cleaning, preservation, packaging, packing, and marking for shipment shall be in accordance with Military Specification MIL-P-15424.

6. NOTES

6.1 Intended use.

6.1.1 Type I, class 1 lifters are suitable for removing and replacing valve springs of L-head, V, and overhead valve engines.

6.1.2 Type I, class 2 lifters are suitable for removing and replacing valve springs of L-head gasoline engines.

6.1.3 Type II, style A compressors are suitable for removing and replacing valve springs for small engines such as Briggs-Stratton and Clinton.

6.1.4 Type II, style B compressors are suitable for removing and replacing valve springs of overhead valve engines and replacing springs in gasoline engines of all types.

6.2 **Ordering data.**—Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and style of lifter and compressor required. (See 1.2.)
(c) Size required (see tables I through VI).
(d) Type of extra offset jaws required. (See 3.9.1.2.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

(e) If valve spring holders are to be furnished. (See 3.9.1.4.)

(f) Whether preparation for immediate use, domestic shipment and storage, or overseas shipment is required (see 5.2.1).

6.2.1 When repair parts are requested, specify the following:

(a) The manufacturer's name or trade-mark of the compressor or lifter for which required.

(b) Quantity and proper name or description of the part required.

(c) Type, class, style, size and manufacturer's number of the lifter and compressor for which the part is required (see 3.6).

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—E O Sig T

Navy—Sh MC Or Y

Air Force.

Copies of this specification may be purchased for 15 cents.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.